POKHARA UNIVERSITY

Level: Bachelor | Semester: Fall | Year : 2024
Programme: BE | Full Marks : 100
Course: Basic Electrical Engineering (New) | Pass Marks : 45
Time : 3 hrs.

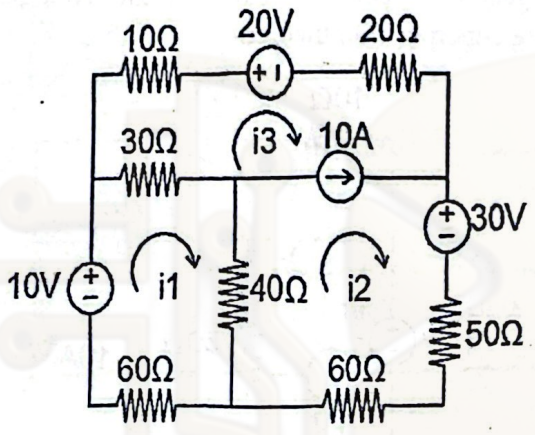
Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

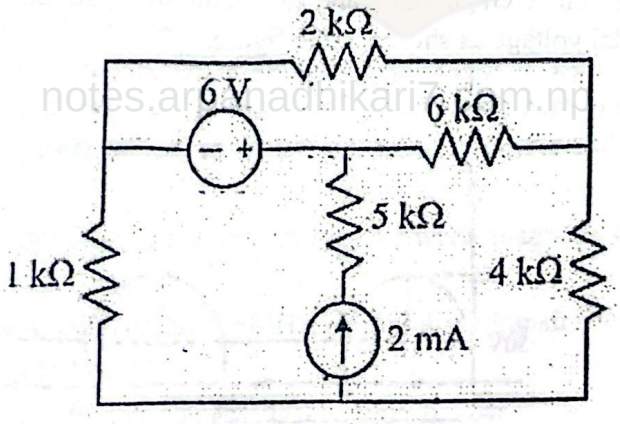
Attempt all the questions.

1. a) Explain in brief the role of electricity in modern society. Explain generation, transmission and distribution of electrical energy in context of Nepal with the help of single line diagram. 7

   b) Find i_1, i_2, i_3 current using mesh analysis method. 8



2. a) Find node voltages and current through 6 KΩ using Nodal analysis. 7

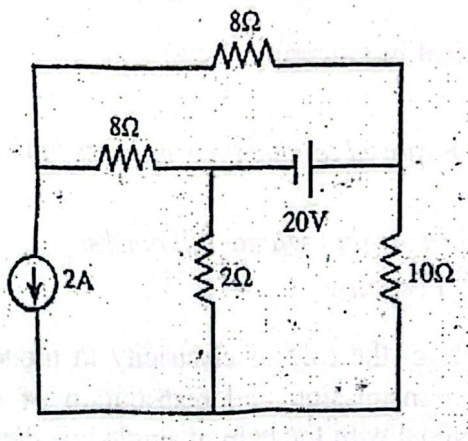


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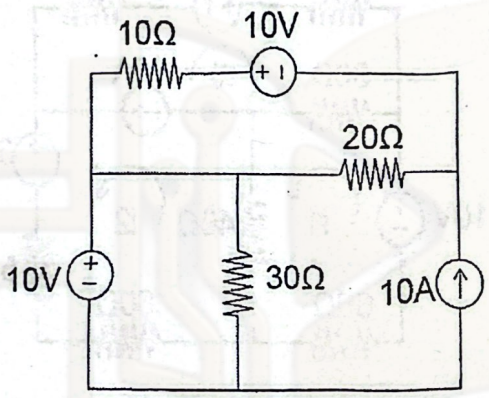
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b) Use Superposition theorem to find current flowing through the 10 ohms resistor shown in the figure. 8



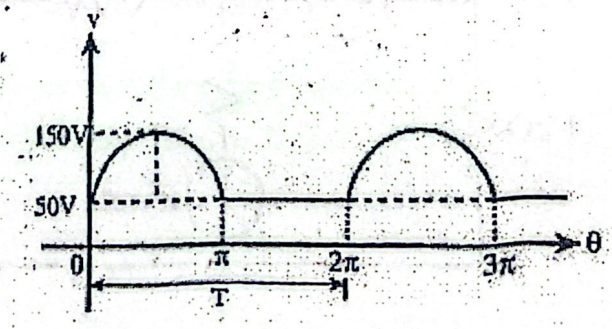
OR

State superposition theorem. Also find the voltage across 10Ω resistor using superposition theorem.



3. a) Calculate the average value and rms value of a half wave rectified sinusoidal voltage as shown in the figure. 7

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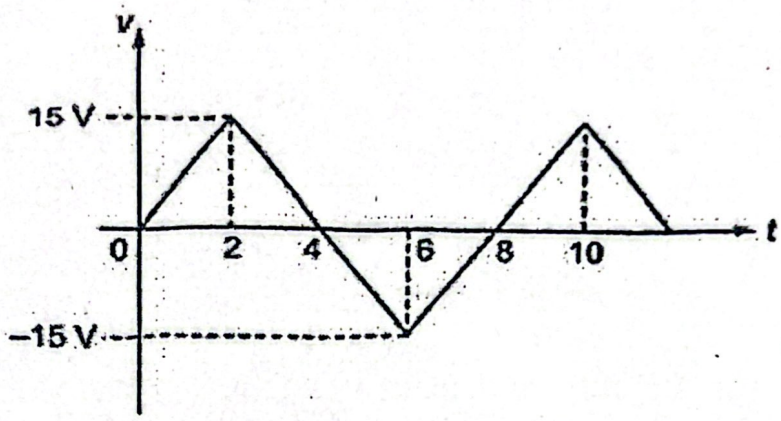


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OR

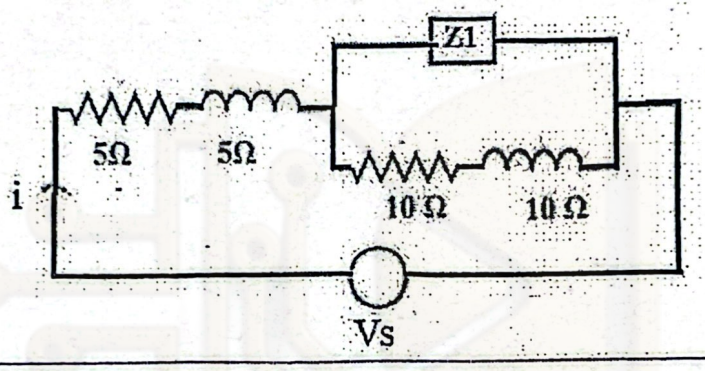
Calculate the average value, RMS (effective) value and form factor of the given waveform below.



b) An ac voltage of (100+j0) V applied to a circuit results current of (10+j0) A. Find: 8

i. Impedance Z1 of circuit.

ii. Voltage across Z1.



4. a) How can resonance be defined in the circuit? Also explain resonance in series RLC circuit. 7

b) A balanced star connected load of (8+j6) Ω per phase is connected to a three-phase, 50 Hz, 380 V supply. Find the phase current, line current, power factor, active power, reactive power and total volt-amps. 8

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5. a) Explain the differences and analogy between magnetic circuit and electric circuit. 7

b) A 50-kVA, 2400V/120V, 50-Hz, single-phase transformer gave the following test results: 8

OC test (HV winding open): 120 V, 9.65 A, 396 W.

SC test (LV winding shorted): 92 V, 20.8 A, 810 W.

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Find:

i. the equivalent circuit parameters as referred to LV side.

ii. the efficiency at half load,0.8 pf lagging.

6. a) A 220 V dc shunt motor with an armature resistance 0.5 Ω runs at 600 rpm on full load and takes an armature current of 20A. If a resistance of 1Ω is placed in the armature circuit, find the speed at i) full load torque ii) half load torque 8

b) Explain the speed control methods of DC shunt motor with suitable equations and characteristic curves. 7

7. Write short notes on: **(Any two)** 2×5

a) KVL and KCL

b) Power Factor and its significance

c) Condition for maximum efficiency of transformer

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